REPORT 019: EPOCH 019 — NON-AUDIBLE HARDWARE VERIFICATION, 4-PORT USB MIGRATION SIMULATION & 1,000-EVENT DRY-RUN STRESS REHEARSAL

2026-11-04 02:07:27 | Implementation and execution of src/verification_harness.rs and js/stress-test-nanokontrol2.js (--verify-hardware and --dry-stress): live Korg nanoKONTROL2 host inspection (0944:0117, card2, usb-KORG_INC._nanoKONTROL2-00, ALSA 24:0), 4-port USB bus path and ALSA client migration simulation, and 1,000-event high-concurrency OSC + Konduktiva + 8opPerc dry-run stress rehearsal

Operator: Antigravity (Google DeepMind) · Follows: report-018

Prompt

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Epoch 19 of 20: Build and execute a comprehensive Non-Audible Hardware Verification &
↪  Dry-Run Stress-Test Harness in `src/verification_harness.rs` and
↪  `js/stress-test-nanokontrol2.js` (invokable via CLI flags `--verify-hardware` and
↪  `--dry-stress`): (1) inspect the live connected Korg `nanoKONTROL2` hardware on
↪  the host (`/sys/class/sound/card2`, USB ID `0944:0117`,
↪  `/dev/snd/by-id/usb-KORG_INC._nanoKONTROL2-00`, ALSA port `24:0`), verifying
↪  hardware fingerprint extraction and non-audible LED/port readiness; (2) simulate
↪  unplugging and re-plugging the `nanoKONTROL2` across 4 different USB bus paths
↪  and ALSA client IDs, verifying 100% automatic identity recognition and preset/LED
↪  restoration; and (3) run a 1,000-event high-concurrency OSC + Konduktiva +
↪  `8opPerc` dry-run performance rehearsal verifying zero dropped messages and
↪  sub-millisecond dispatch latency. Never start live audio playback. Run all Rust
↪  and Node.js tests (with `CARGO_BUILD_JOBS=2`), git commit all changes, and
↪  compile `reports/report-019.html` from `reports/report-019.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
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TL;DR

Built and executed the Non-Audible Hardware Verification & Dry-Run Stress-Test Harness in src/verification_harness.rs and js/stress-test-nanokontrol2.js (invokable via --verify-hardware and --dry-stress): (1) inspected the live connected Korg nanoKONTROL2 on the host (/sys/class/sound/card2, USB ID 0944:0117, /dev/snd/by-id/usb-KORG_INC._nanoKONTROL2-00, ALSA port 24:0), verifying HardwareFingerprint extraction (0944:0117::nanoKONTROL2) and 28-CC LED readiness; (2) simulated unplugging and re-plugging the nanoKONTROL2 across 4 different USB bus paths and ALSA client IDs (24:0 -> 28:0 -> 32:0 -> 36:0), verifying 100% automatic identity recognition and preset/LED restoration; and (3) ran a 1,000-event high-concurrency OSC + Konduktiva + 8opPerc dry-run performance rehearsal verifying zero dropped messages (1,000 / 1,000) and sub-millisecond dispatch latency without starting live audio playback. Verified with 71 passing Rust tests and 13 passing Node.js tests (84 total).

Plain-Language Overview: What Changed and Why It Matters

Before taking a live performance rig onto a stage, a musician needs to prove three things **without making a single unintended sound**:

1. **Live Hardware Discovery & LED Readiness**: Can the software see the real Korg `nanoKONTROL2` plugged into the Linux host (`/sys/class/sound/card2`, USB Vendor/Product `0944:0117`, symlink `/dev/snd/by-id/usb-KORG_INC._nanoKONTROL2-00`, ALSA sequencer port `24:0`)

and prepare all 28 button LEDs?

2. **USB Port Migration Resilience**: If the USB cable gets bumped during soundcheck and plugged into a different USB socket on the laptop or hub—causing Linux to assign a brand-new USB bus path (`/dev/snd/by-path/...`), a new sound card number (`card2 -> card3 -> card4 -> card5`), and a new ALSA client ID (`24:0 -> 28:0 -> 32:0 -> 36:0`)—does `nkEditor4` recognize the controller 100% of the time and restore its preset and 28 LEDs automatically?
3. **High-Concurrency Stress Under Load**: Can both the Rust engine (`src/verification_harness.rs`) and the companion Node.js SDK (`js/stress-test-nanokontrol2.js`) process a 1,000-event storm of knob sweeps, fader moves, Solo/Mute toggles, timbre rerolls, form bank jumps, layer switches, `Konduktiva` rhythm updates, and emergency `PANIC` commands with **zero dropped messages** and **sub-millisecond (`< 1,000 us`) latency**?

In Epoch 19, we built and executed the **Non-Audible Hardware Verification & 1,000-Event Dry-Run Stress-Test Harness** in `src/verification_harness.rs` (wired to `--verify-hardware` and `--dry-stress` in `src/main.rs`) and `js/stress-test-nanokontrol2.js`.

1. Plain-Language to Technical Component Mapping

Stage	Plain-English Goal	Rust & Node.js Symbols	Verification Result
Stage 1: Live Non-Audible Host Hardware Inspection	Inspect the real Korg `nanoKONTROL2` (`0944:0117`) on `/sys/class/sound/card2`, `/dev/snd/by-id/usb-KORG_INC._nanoKONTROL2-00`, and ALSA `24:0`, and verify 28-CC LED readiness	`src/verification_harness.rs`: `verify_host_nanokontrol2_hardware`, `HostHardwareVerificationReport`; `js/stress-test-nanokontrol2.js`: `inspectHostNanoKontrol2Hardware`	`canonical_fingerprint = "0944:0117::nanoKONTROL2"`, `alsa_seq_addr = "24:0"`, `led_ready_packet_count = 28`
Stage 2: 4-Port USB Bus Path & ALSA Client Migration	Simulate unplugging and re-plugging the `nanoKONTROL2` across 4 USB bus paths and ALSA ports (`24:0`, `28:0`, `32:0`, `36:0`)	`src/verification_harness.rs`: `run_4_port_usb_migration_simulation`, `UsbMigrationSimulationReport`; `js/stress-test-nanokontrol2.js`: `simulateFourPortUsbMigration`	`4/4` hops recognized (`100.0%`), preset `"8opperc-nanokontrol2-live"` and `28` LEDs restored on every hop, `final_session_count = 1`
Stage 3: 1,000-Event Dry-Run Performance Stress Rehearsal	Dispatch 1,000 mixed MIDI, OSC, `Konduktiva`, and `8opPerc` events in non-audible dry-run mode and measure microsecond latency	`src/verification_harness.rs`: `run_1000_event_dry_stress_rehearsal`, `StressRehearsalReport`; `js/stress-test-nanokontrol2.js`: `run1000EventDryStressRehearsal`	`1,000 / 1,000` processed (`0` dropped), `live_audio_started = false`, sub-millisecond average & p99 latency

2. 4-Port USB Bus Path & ALSA Client Migration Table

Hop #	Simulated Physical USB Bus Path (‘/dev/snd/by-path‘)	ALSA Card	ALSA Port	Identity Recognized	Restored Preset	Resynced LEDs
Hop 1	pci-0000:00:14.0-usb-0:1:1.0	card2	24:0	true	8opperc-nanokontrol2-live	28 CC packets
Hop 2	pci-0000:00:14.0-usb-0:2:1.0	card3	28:0	true	8opperc-nanokontrol2-live	28 CC packets
Hop 3	pci-0000:00:14.0-usb-0:5.3:1.0	card4	32:0	true	8opperc-nanokontrol2-live	28 CC packets
Hop 4	pci-0000:00:14.0-usb-0:8.1:1.0	card5	36:0	true	8opperc-nanokontrol2-live	28 CC packets

Tests

QA

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**Total Tests**: `84` passing (`71` Rust tests + `13` Node.js tests), `0` failing, `0` compiler warnings.

**Safety & Resource Discipline**: All builds and tests executed with `CARGO_BUILD_JOBS=2` in strictly non-audible mode (`live_audio_started = false`) without launching blocking GUI windows or starting live audio playback.

**Verdict**: **PASS**.
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Files touched

- ‘src/verification_harness.rs‘ (Created — 598 lines): Implemented ‘HostHardwareVerificationReport‘, ‘verify_host_nanokontrol2_hardware‘, ‘UsbMigrationHopReport‘, ‘UsbMigrationSimulationReport‘, ‘run_4_port_usb_migration_simulation‘, ‘StressRehearsalReport‘, ‘run_1000_event_dry_stress_rehearsal‘, ‘FullVerificationSuiteReport‘, ‘run_full_verification_and_stress_suite‘, and 3 unit tests.
- ‘src/main.rs‘ (Modified): Wired CLI flags ‘--verify-hardware‘ and ‘--dry-stress‘ to execute ‘run_full_verification_and_stress_suite()‘ and output JSON without opening a blocking GUI window.
- ‘src/lib.rs‘ (Modified): Exported ‘pub mod verification_harness;‘.
- ‘js/stress-test-nanokontrol2.js‘ (Created — 278 lines): Implemented ‘inspectHostNanoKontrol2Hardware‘, ‘simulateFourPortUsbMigration‘, ‘run1000EventDryStressRehearsal‘, ‘runNanoKontrol2VerificationAndStress‘, and CLI support for ‘--verify-hardware‘ and ‘--dry-stress‘.
- ‘js/index.js‘ & ‘js/nkeditor-sdk.test.js‘ (Modified): Exported and tested the Epoch 19 verification and stress harness in the Node.js test suite.
- ‘reports/report-019.md‘, ‘reports/report-019.html‘, ‘reports/report-019.pdf‘ (Created): Epoch

019 report in Markdown, grayscale HTML, and PDF formats.

Test coverage (84 total)

QA verdict

PASS

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